

Vatsal Malkari

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EDUCATION

Rutgers University

Master of Science in Data Science

New Brunswick, NJ

Expected May 2027

- **Coursework:** Data Structures & Algorithms, Statistical Learning in Data Science (Deep Learning), Statistical Software (NLP, LLMs, Transformers), Special Topics in Data Science (Databases), Probability & Inference, Data Mining with R

Rutgers University

Bachelor of Science in Computer Science and Data Science

New Brunswick, NJ

Sep 2022 – May 2025

- **Coursework:** Data Structures & Algorithms, Systems Programming, Computer Architecture, Database Management, Deep Learning, Regression & Statistical Methods

EXPERIENCE

Nokia Bell Labs

Software Verification Engineer Co-op

May 2026 – Aug 2026

- Reverse engineered REST API specs with cross functional teams to build an isolated simulator, enabling safe testing without risking production hardware
- Scripted Python and Shell tools to parse REST JSON payloads for node setup and IP allocation, reducing manual setup time from 4+ hours to under 30 minutes
- Engineered helper tools and workflow pipelines enabling AI agents to execute network creation, link updates, and fault injections via natural language
- Containerized backend automation logic with Docker, for reproducible test execution and developer onboarding

New Jersey Cybersecurity and Communications Integration Cell

Data Engineering Extern

Jan 2025 – May 2025

via Rutgers MBS Externship Exchange

- Built a scalable pipeline on Google Cloud Platform processing 2M+ weekly login events to identify top 1% high-risk accounts
- Implemented anomaly detection in BigQuery ML, improving threat detection accuracy by 22%
- Automated threat classification using PCA and vectorized workflows, reducing analyst time

PROJECTS

PSS Telecom Network Simulator

Jun 2026 – Sep 2026

- Built a C++ discrete event engine processing 1.36M events per second using Dijkstra routing, 1+1 protection switching, and lock-free concurrency with zero race conditions
- Deployed a Flask REST API, SQLite backend, and C++ engine using Docker and Kubernetes, scaling up to 10K services within a 43.8MB memory footprint

Custom C++ Deep Learning Engine

Jan 2026 – Sep 2026

- Built a framework-free engine with a custom DAG Autograd graph for automated backpropagation
- Designed a bitwise-aligned Memory Arena, reducing OS heap allocation to <0.1% of thread runtime
- Optimized Conv2D, ReLU, Flatten layers via pointer tracking, reaching 44.2% accuracy with zero memory leaks
- Deployed real-time webcam inference (<100 ms) via a hybrid C++ (-O3 -fast-math) and Python/OpenCV pipeline

Python Interpreter in C

Nov 2025 – Present

- Implemented core compiler components (tokenizer, parser, evaluator) with security first design: bounds checking on all buffers, reference counting, input validation. Zero buffer overflows, zero memory leaks across 40+ tests.
- Debugged parser bug in nested blocks using AddressSanitizer; identified and fixed state management issue affecting control flow. Demonstrated proficiency with compiler correctness and memory safety tools.
- Demonstrated passion for eliminating vulnerabilities at language level through secure C design principles: safe error handling for edge cases, input validation, and defensive coding practices.

Low-Level Task Scheduler

Jul 2025 – Feb 2026

- Engineered a C scheduler using Min Heap and Hash Map for $O(\log N)$ scheduling and $O(1)$ cancellation
- Replaced busy-waiting with pthread condition variables, improving CPU efficiency and thread safety
- Stress-tested 2,000 concurrent tasks, resolving multithreaded deadlocks

SKILLS

Languages: C++, Python, Java, SQL, Shell, C

Systems: Linux, Multithreading, Lock-free Programming, Memory Management, Smart Pointers, Thread Safety

Machine Learning & Data: TensorFlow, PyTorch, Scikit-learn, BigQuery ML, NLP, LLMs, Deep Learning, Spark

Backend: REST APIs, Distributed Systems, Discrete Event Simulation, GCP, Docker, Kubernetes, Git, CI/CD